AI Hallucinations in Legal Cases

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Research Question, Exploratory Analysis, and Decision Tree

Research Question

Artificial intelligence tools are increasingly being used in legal settings, but AI-generated hallucinations can introduce inaccurate information into legal documents and proceedings. Our research question is:

What characteristics of legal cases involving AI hallucinations are associated with whether a professional sanction is issued?

We focus on characteristics of the parties involved and the primary legal field of the case. Our response variable is whether the case resulted in a professional sanction.

Data Preparation

```
library(tidyverse)
```

```
-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v dplyr     1.2.1     v readr     2.1.5
v forcats   1.0.0     v stringr   1.5.1
v ggplot2   4.0.3     v tibble    3.3.0
v lubridate 1.9.4     v tidyr     1.3.2
v purrr     1.2.2
-- Conflicts ------------------------------------------ tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to becor
```

```
library(rpart)
library(rpart.plot)

hallucination <- read_csv("hallucination_cases.csv")
```

```
Rows: 2078 Columns: 18
-- Column specification --------------------------------------------------------
Delimiter: ","
chr  (17): Case Name, Court, State(s), Party(ies), AI Tool, Hallucination It...
date  (1): Date

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
hallucination <- hallucination |>
  mutate(
    sanction = factor(
      `Professional Sanction`,
      levels = c("No", "Yes")
    ),

    # Group party types for exploratory analysis
    party_group = case_when(
      `Party(ies)` == "Lawyer" ~ "Lawyer",
      `Party(ies)` == "Pro Se Litigant" ~ "Pro Se Litigant",
      TRUE ~ "Other"
    ),

    # Variables used in the decision tree
    party = factor(`Party(ies)`),
    legal_field = factor(`Legal Field Primary`),
    alleged = factor(Alleged),
    vendor_disputed = factor(`Vendor Disputed`)
  ) |>
  drop_na(sanction)
```

Exploratory Analysis

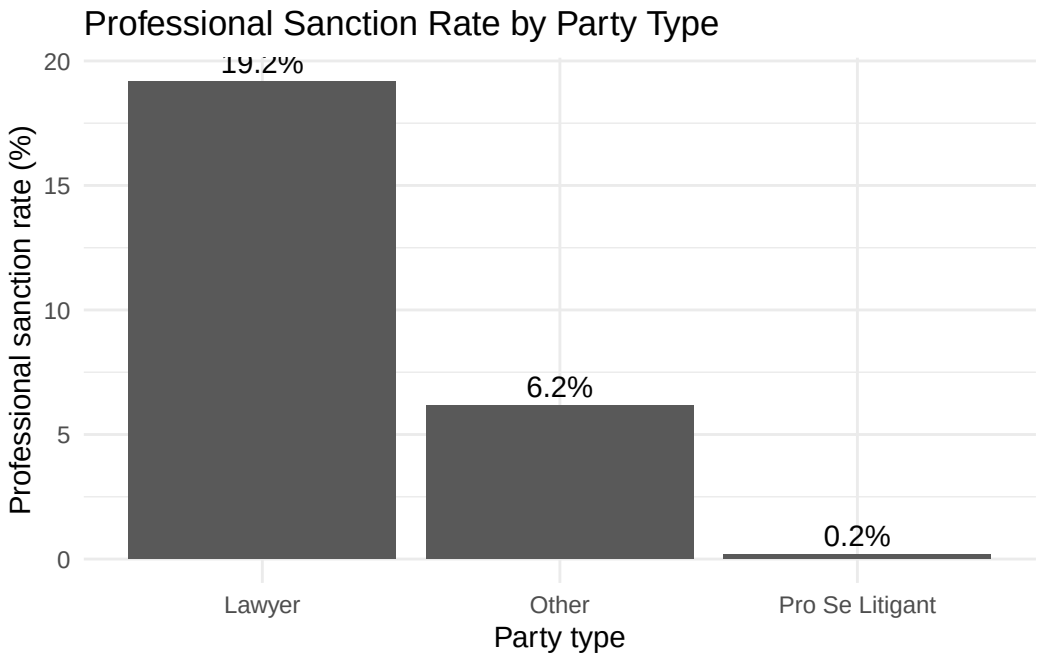
Because our response variable is whether a professional sanction was issued, we first examine whether the sanction rate differs across party types.

```r
party_summary <- hallucination |>
  group_by(party_group) |>
  summarise(
    cases = n(),
    sanctions = sum(sanction == "Yes"),
    sanction_rate = mean(sanction == "Yes") * 100
  )

party_summary
```

```
# A tibble: 3 x 4
  party_group     cases sanctions sanction_rate
  <chr>           <int>     <int>         <dbl>
1 Lawyer            819       157        19.2
2 Other              65         4         6.15
3 Pro Se Litigant  1194         2         0.168
```

```r
ggplot(
  party_summary,
  aes(x = party_group, y = sanction_rate)
) +
  geom_col() +
  geom_text(
    aes(label = paste0(round(sanction_rate, 1), "%")),
    vjust = -0.4
  ) +
  labs(
    title = "Professional Sanction Rate by Party Type",
    x = "Party type",
    y = "Professional sanction rate (%)"
  ) +
  theme_minimal()
```



The exploratory analysis shows substantial differences in professional sanction rates across party types. Cases involving lawyers have a considerably higher sanction rate than cases involving pro se litigants, while cases classified as other party types fall between these groups. This suggests that party type may be an important predictor of professional sanctions and motivates its inclusion in the decision tree.

Decision Tree

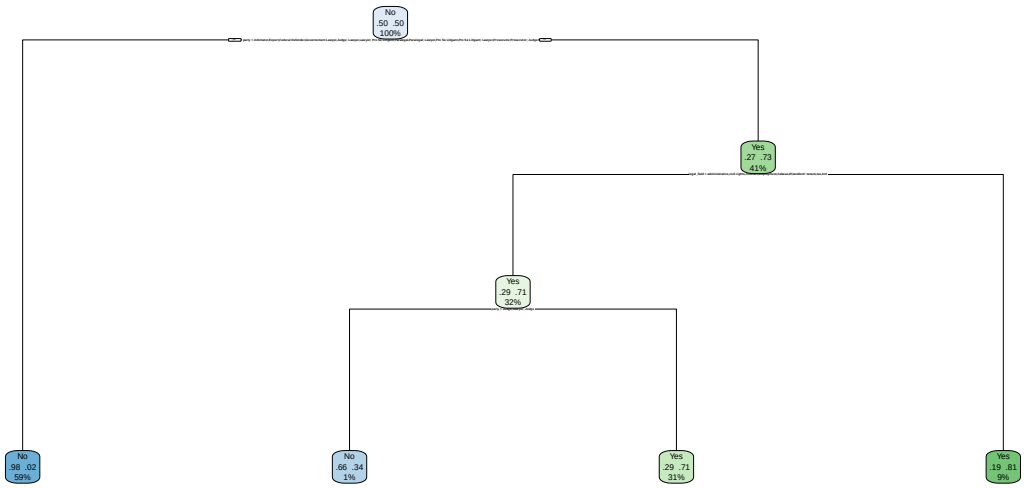
To examine whether party type and other case characteristics jointly predict professional sanctions, we fit a classification tree using party type, primary legal field, alleged misconduct, and whether the AI vendor disputed the hallucination.

Because professional sanctions are relatively uncommon in the dataset, we use equal class priors so that the tree does not simply favor the majority class of cases with no sanction.

```
tree <- rpart(
  sanction ~ party + legal_field + alleged + vendor_disputed,
  data = hallucination,
  method = "class",
  parms = list(prior = c(0.5, 0.5)),
  control = rpart.control(
    cp = 0.001,
    minsplit = 20,
```

```
    minbucket = 10
  )
)
```

```
rpart.plot(
  tree,
  type = 2,
  extra = 104,
  fallen.leaves = TRUE,
  faclen = 0,
  varlen = 0,
  split.cex = 0.35,
  cex = 0.7,
  compress = FALSE
)
```




```
rpart.rules(
  tree,
  style = "tall",
  cover = TRUE
)
```

```
sanction is 0.02 with cover 59% when
    party is Arbitrator or Expert or Federal Defender or Governement Lawyer or Judge; Lawyer
```

```
sanction is 0.34 with cover 1% when
    party is Judge or Lawyer; Judge
    legal_field is administrative or civil rights or contract or employment or habeas or IP
tenant or tax or tort

sanction is 0.71 with cover 31% when
    party is Lawyer
    legal_field is administrative or civil rights or contract or employment or habeas or IP
tenant or tax or tort

sanction is 0.81 with cover 9% when
    party is Judge or Judge; Prosecutor or Lawyer or Lawyer; Judge
    legal_field is bankruptcy or criminal or family or immigration or other
```

The decision tree identifies party type and legal field as important predictors of professional sanctions. The first split is based on party type, indicating that this variable provides the greatest reduction in classification error among the predictors considered. One branch contains approximately 59% of the cases and has a very low model-assigned probability of a sanction.

Among the remaining cases, legal field provides an additional important split. The tree identifies combinations of party type and legal field associated with substantially higher predicted probabilities of professional sanctions. In particular, one terminal node has a model-assigned probability of approximately 81% for a professional sanction and contains approximately 9% of the cases.

Overall, the tree suggests that professional sanctions are not evenly distributed across cases involving AI hallucinations. Instead, the predicted likelihood of a sanction varies substantially depending on the characteristics of the parties involved and the legal field of the case.